Chapter 5

Vibrations of MDOF systems

5.1 Equation of motion of mechanical systems

5.1.1 Dynamical systems and mechanical systems

A dynamical system is described by an equation in the form:

$$\dot{\mathbf{x}} = \mathbf{f}\left(\mathbf{x}, t\right) \ , \ \mathbf{x}\left(t\right) \in \mathbb{R}^n \tag{5.1}$$

Where $\mathbf{f}\left(\mathbf{x}, t\right)$ is the *vector field* and $\mathbb{R}^n$ is the *state space*. In this form, the system is non-autonomous, while an autonomous system is:

$$\dot{\mathbf{x}} = \mathbf{f}\left(\mathbf{x}\right) \ , \ \mathbf{x}\left(t\right) \in \mathbb{R}^n \tag{5.2}$$

The evolution of the system is a trajectory in the state space starting from the initial conditions:

$$\mathbf{x}\left(0\right) = \mathbf{x_0} \tag{5.3}$$

A mechanical system is a dynamical system governed by Lagrange's equations:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_k}\right) + \frac{\partial T}{\partial q_k} = Q_k \ \ , \ \ k = 1, \ldots n \tag{5.4}$$

Where T is the kinetic energy and $q_k \in \mathbb{R}^n$ is the $k - th$ *generalized coordinate*; $\mathbb{R}^n$ is the *configuration space*.
The vector Q_k is the $k - th$ *generalized force*:

$$Q_n = \sum_{s=1}^{N} \mathbf{F_s} \cdot \frac{\partial \mathbf{P_s}}{\partial q_n} \tag{5.5}$$

where $\mathbf{P_s}$ is the point at which the force $\mathbf{F_s}$ is applied.
If the system has holonomic constraints, then T is a second order polynomial of the generalized velocities $\dot{q}_k$. If the constraints are also scleronomous, then T is a quadratic form of the generalized velocities:

$$T = \frac{1}{2}\sum_{i,j=1}^{n} m_{ij}\left(\mathbf{q}\right)\dot{q}_i\dot{q}_j = \frac{1}{2}\dot{\mathbf{q}}^T\mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}} = \frac{1}{2}\left(\mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}}\right)\cdot\dot{\mathbf{q}} \tag{5.6}$$

M is positive definite symmetric matrix, *ie.*:

$$T = \frac{1}{2}\dot{\mathbf{q}}^T\mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}} > 0 \ \ , \ \ \forall\mathbf{q} \neq 0 \tag{5.7}$$

If the constraints are preventing all the rigid body motion, then $\mathbf{M}(\mathbf{q})$ is a constant matrix $\mathbf{M}(\mathbf{q}) = \mathbf{M}$. In this case, the equation of motion is:

$$\mathbf{M}\ddot{\mathbf{q}} = \mathbf{f}^{int}\left(\mathbf{q},\dot{\mathbf{q}}\right) + \mathbf{f}^{ext}\left(\mathbf{q},\dot{\mathbf{q}},t\right) \tag{5.8}$$

If $\mathbf{q} = \mathbf{q_E}$ is an equilibrium point:

$$\mathbf{f}^{int}\left(\mathbf{q_E},0\right) + \mathbf{f}^{ext}\left(\mathbf{q_E},0,t\right) \ \ , \ \ \forall t \tag{5.9}$$

The equation of motion can be linearized in the neighborhood of the equilibrium point:

$$\mathbf{M}\ddot{\mathbf{q}} = \left(\mathbf{C_s^E} + \mathbf{C_g^E}\right)\dot{\mathbf{q}} + \left(\mathbf{K_e^E} + \mathbf{K_g^E}\right)\left(\mathbf{q} - \mathbf{q_E}\right) = \mathbf{0} \tag{5.10}$$

The total stiffness matrix is the summation of the elastic stiffness and of the geometric stiffness:

$$\mathbf{K^E} = \mathbf{K_e^E} + \mathbf{K_g^E} = -\left[\frac{\partial\mathbf{f}^{(int)}}{\partial\mathbf{q}}\right]_E - \left[\frac{\partial\mathbf{f}^{(ext)}}{\partial\mathbf{q}}\right]_E \tag{5.11}$$

The total damping matrix is the summation of the structural damping and of the geometric damping:

$$\mathbf{C^E} = \mathbf{C_s^E} + \mathbf{C_g^E} = -\left[\frac{\partial\mathbf{f}^{(int)}}{\partial\dot{\mathbf{q}}}\right]_E - \left[\frac{\partial\mathbf{f}^{(ext)}}{\partial\dot{\mathbf{q}}}\right]_E \tag{5.12}$$

$\mathbf{K_e^E}$, $\mathbf{C_s^E}$ are symmetric and positive definite. The quadratic form of $\mathbf{K_e^E}$ is the *elastic potential energy* :

$$U = \frac{1}{2}\mathbf{q}^T\mathbf{K_e^E}\mathbf{q} \geq 0 \tag{5.13}$$

The quadratic form of $\mathbf{C_s^E}$ is the *dissipated energy*:

$$E_d = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{C_s^E} \dot{\mathbf{q}} \geq 0 \tag{5.14}$$

$\mathbf{K_g^E}$, $\mathbf{C_g^E}$ are non symmetric and indefinite.
The free vibration problem around the equilibrium point $\mathbf{q_E}$ can be written in the following general formulation:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0} \tag{5.15}$$

5.1.2 Order reduction

The equation of the free vibrations of a mechanical system is a set of n 2nd order equations: this set can be reduced to a set of 2n 1st order equations in the form:

$$\dot{\mathbf{x}} = \mathbf{J}\mathbf{x}\ , \quad \mathbf{x} = \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix}\ , \quad \mathbf{x} \in \mathbb{R}^{2n} \tag{5.16}$$

where $\mathbf{J}$ is the matrix:

$$\mathbf{J} = \begin{bmatrix} 0 & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \tag{5.17}$$

The equation (5.16) is written in the *state space*.

5.2 Undamped MDOF systems

5.2.1 An example of a 2-DOF system: 1/4 car model

A simple model for describing the vertical dynamics of a car is a 2-DOF model in which it is assumed that all the four wheels are vibrating with the same law. In the 1/4 car model, m_s is 1/4 of the whole suspended mass of the vehicle, m_u is 1/4 of the whole unsuspended mass of the vehicle, k_u is the stiffness of one tyre and k_s the stiffness of one suspension. Usually $m_s \gg m_u$ and $k_u \gg k_s$. Tyre damping is negligible with respect to suspension damper, where a shock absorber is used. The system is forced due to the ground asperities $y(t)$, which can be written as an external force $k_u y(t)$.

$$\begin{cases} m_s\ddot{x}_s + c_s\left(\dot{x}_s - \dot{x}_u\right) + k_s\left(x_s - x_u\right) = 0 \\ \\ m_u\ddot{x}_u - c_s\left(\dot{x}_s - \dot{x}_u\right) - k_s\left(x_s - x_u\right) + k_u x_u = k_u y \end{cases} \tag{5.18}$$

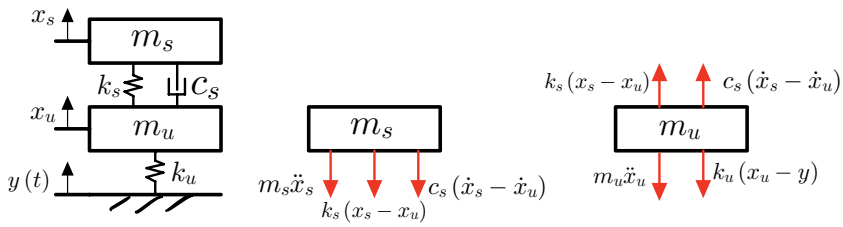

Figure 5.1: 1/4 car model

We would like to write the equations in matrix form, according to the following representation:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{f} \tag{5.19}$$

where $\mathbf{M}$ is a symmetric positive definite matrix, while $\mathbf{K}$ is a symmetric semi-definite positive matrix. For the 1/4 car model, eq. (5.19) becomes:

$$\begin{bmatrix} m_s & 0 \\ 0 & m_u \end{bmatrix} \begin{bmatrix} \ddot{x}_s \\ \ddot{x}_u \end{bmatrix} + \begin{bmatrix} c_s & -c_s \\ -c_s & c_s \end{bmatrix} \begin{bmatrix} x_s \\ x_u \end{bmatrix} + \begin{bmatrix} k_s & -k_s \\ -k_s & k_s + k_u \end{bmatrix} \begin{bmatrix} x_s \\ x_u \end{bmatrix} = \begin{bmatrix} 0 \\ k_u y \end{bmatrix} \tag{5.20}$$

5.2.2 Normal modes of a MDOF system

Considering the free problem:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0} \tag{5.21}$$

we look for the *synchronous motions*, *i.e.* solutions in the form $\mathbf{q}(t) = \mathbf{u}f(t)$, where $\mathbf{u} \in \mathbb{R}^n$ is a constant vector and $f(t)$ is a scalar function of time.
We replace this solution into the equation of the free problem and then we multiply on the left by $\mathbf{u}^T$, so that we obtain a scalar equation:

$$\begin{gathered} \mathbf{M}\mathbf{u}\ddot{f}(t) + \mathbf{K}\mathbf{u}f(t) = \mathbf{0} \\ \mathbf{u}^T\mathbf{M}\mathbf{u}\ddot{f}(t) + \mathbf{u}^T\mathbf{K}\mathbf{u}f(t) = 0 \end{gathered} \tag{5.22}$$

We divide by $\mathbf{u}^T\mathbf{M}\mathbf{u}$, which is positive for any $\mathbf{u}$:

$$-\frac{\mathbf{u}^T\mathbf{K}\mathbf{u}}{\mathbf{u}^T\mathbf{M}\mathbf{u}} = \frac{\ddot{f}(t)}{f(t)} = -\lambda = -\omega^2 \tag{5.23}$$

therefore the ratio $\dfrac{\ddot{f}(t)}{f(t)}$ must be constant; we will have to prove that it is a negative constant $-\omega^2$. Two equations arise from the last relationship:

$$\begin{cases} \ddot{f}(t) + \omega^2 f(t) = 0 \\ \mathbf{u}^T\mathbf{K}\mathbf{u} + \omega^2\mathbf{u}^T\mathbf{M}\mathbf{u} = 0 \end{cases} \tag{5.24}$$

the first equation is a 1-DOF oscillator, so we need to find the couples ($\mathbf{u}$, ω^2) such that the second equation holds. Replacing $\ddot{f} = -\omega^2 f$ in the first of the (5.22) and recalling that $f(t) = 0$ is a trivial solution one gets:

$$\left(-\omega^2\mathbf{M} + \mathbf{K}\right)\mathbf{u} = \mathbf{0} \tag{5.25}$$

or equivalently, premultiplying by $\mathbf{M}^{-1}$:

$$\begin{gathered} \left(-\omega^2\mathbf{I} + \mathbf{M}^{-1}\mathbf{K}\right)\mathbf{u} = \mathbf{0} \\ \left(\mathbf{M}^{-1}\mathbf{K}\right)\mathbf{u} = \omega^2\mathbf{u} \end{gathered} \tag{5.26}$$

Eq. (5.26) holds for the eivenvalues ω^2 and the eigenvectors $\mathbf{u}$ of the matrix $(\mathbf{M}^{-1}\mathbf{K})$; eq. (5.25) is a generalized eigenvalue problem of $\mathbf{M}$ and $\mathbf{K}$. Both approaches are acceptable to define the eigenvalue problem (the second one can be solved faster using numerical methods). The eigenvalues are found by solving the *characteristic equation*:

$$\det\left(-\omega^2\mathbf{M} + \mathbf{K}\right) = 0 \tag{5.27}$$

5.2.3 Natural frequencies and poles of a MDOF system

Another possible approach is based on Laplace's transform of eq. (5.21):

$$\begin{gathered} \left(s^2\mathbf{M} + \mathbf{K}\right)\mathbf{Q}(s) = 0 \\ \left(\mathbf{A} + s^2\mathbf{I}\right)\mathbf{Q}(s) = 0 \end{gathered} \tag{5.28}$$

where $\mathbf{A} = \mathbf{M}^{-1}\mathbf{K}$ has eigenvalues $\lambda = -s^2 = \omega^2$, therefore:

$$s = i\omega = i\left(\pm\sqrt{\lambda}\right) = \pm\sqrt{-\lambda} \tag{5.29}$$

where s are the *poles* of the system, ω are the *natural circular frequencies* and they will be indicated with ω_n in the following. The corresponding *natural frequencies* are $f_n = \frac{\omega_n}{2\pi}$. In the example of the 1/4 car model, the matrix $\mathbf{A}$ has size 2x2, therefore 2 eigenvalues and 2 natural frequencies can be found. In general, a MDOF system has n natural frequencies.

5.2.4 Normal modes of the 1/4 car

Considering eq. (5.20), the problem can be made simpler assuming $m_s = 10m_u = 10m$ and $k_u = 8k_s = 8k$. These assumptions describe the most common situation: unsuspended mass is much smaller than suspended mass and tyre stiffness is much larger than suspension stiffness. Possible numerical data are summarized in the tab. 5.1.

parameter	value
m_s	$\frac{1000}{4}kg = 250kg$
k_s	$\frac{70}{4}\frac{kN}{m} = 17.5\frac{kN}{m}$
m_u	$\frac{100}{4}kg = 25kg$
k_u	$\frac{560}{4}\frac{kN}{m} = 140\frac{kN}{m}$

Table 5.1: Numerical values of the parameters of the 1/4 car model

With the assumptions made, the matrixes $\mathbf{M}$ and $\mathbf{K}$ become:

$$\mathbf{M} = \begin{bmatrix} 10m & 0 \\ 0 & m \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} k & -k \\ -k & 9k \end{bmatrix} \tag{5.30}$$

the characteristic equation is $\det(\mathbf{K} - \omega^2\mathbf{M}) = 0$:

$$\begin{vmatrix} k - 10m\omega^2 & -k \\ -k & 9k - m\omega^2 \end{vmatrix} = 0$$

$$10m^2\omega^4 - 91mk\omega^2 + 8k^2 = 0 \tag{5.31}$$

$$\omega_{1,2}^2 = \frac{91km \pm \sqrt{91^2 - 320}km}{20m^2} = \frac{91 \pm \sqrt{7961}}{20}\frac{k}{m}$$

the corresponding natural frequencies, for the given values of the parameters, are:

$$\begin{cases} f_1 = \dfrac{\omega_1}{2\pi} = 1.25Hz \\ \\ f_2 = \dfrac{\omega_2}{2\pi} = 12.64Hz \end{cases} \tag{5.32}$$

To find the eigenvector of the first natural frequency, we will replace ω_1^2 (the smallest eigenvalue) into $(\mathbf{K} - \omega_1^2\mathbf{M})\,\mathbf{u}^{(1)} = 0$:

$$\begin{bmatrix} k - 10m\dfrac{91-\sqrt{7961}}{20}\dfrac{k}{m} & -k \\ -k & 9k - m\dfrac{91-\sqrt{7961}}{20}\dfrac{k}{m} \end{bmatrix} \begin{bmatrix} u_1^{(1)} \\ u_2^{(1)} \end{bmatrix} = 0$$

(5.33)

$$\begin{bmatrix} \dfrac{-89+\sqrt{7961}}{2} & -1 \\ -1 & \dfrac{-89+\sqrt{7961}}{20} \end{bmatrix} \begin{bmatrix} u_1^{(1)} \\ 1 \end{bmatrix} = 0$$

where $u_2^{(1)}$ has been arbitrarily set to 1, since the two equations are linearly dependent upon each other (the matrix is singular after replacing $\omega^2 = \omega_1^2$. The first, or equivalently the second, equation gives:

$$u_1^{(1)} = \frac{2}{-89+\sqrt{7961}} \quad \Rightarrow \quad u^{(1)} = \begin{bmatrix} \dfrac{2}{-89+\sqrt{7961}} \\ 1 \end{bmatrix} \cong \begin{bmatrix} 8.9112 \\ 1 \end{bmatrix} \tag{5.34}$$

Therefore $u^{(1)}$ is the eigenvector of the first mode: the first mode is affecting the ride comfort of the vehicle, and its frequency is the *ride frequency*. If the excitation due to the road profile has harmonic content at frequencies close to $f_1 = 1.25Hz$, the vehicle frame has large oscillations.
The eigenvector of the second mode is found by replacing $\omega^2 = \omega_2^2$ (the largest eigenvalue):

$$\begin{bmatrix} k - 10m\dfrac{91+\sqrt{7961}}{20}\dfrac{k}{m} & -k \\ -k & 9k - m\dfrac{91+\sqrt{7961}}{20}\dfrac{k}{m} \end{bmatrix} \begin{bmatrix} u_1^{(2)} \\ u_2^{(2)} \end{bmatrix} = 0$$

(5.35)

$$\begin{bmatrix} \dfrac{-89-\sqrt{7961}}{2} & -1 \\ -1 & \dfrac{89-\sqrt{7961}}{20} \end{bmatrix} \begin{bmatrix} u_1^{(2)} \\ 1 \end{bmatrix} = 0$$

again, $u_2^{(2)}$ has been arbitrarily set to 1 since the matrix is singular. From one of

the two equations:

$$\mathbf{u}^{(2)} = \begin{bmatrix} \dfrac{2}{-\sqrt{7961}-89} \\ \\ 1 \end{bmatrix} \cong \begin{bmatrix} -0.01122 \\ \\ 1 \end{bmatrix} \tag{5.36}$$

The second mode has x_s out of phase with respect to x_u and x_u is much larger than x_s. This mode is affecting road holding of the vehicle: if the excitation due to the road profile has harmonic content at frequencies close to $f_2 = 12.64Hz$, the dynamic force between the wheel and the ground can be much lower than the static value, so that the vehicle looses traction. The force at the ground is:

$$F_y(t) = F_{yst} + k_u \left(x_u(t) - y(t)\right) \tag{5.37}$$

where F_{yst} is the static force, while $F_y(t)$ is the dynamic force, which can be reduced if $x_u(t)$ has large oscillations. The available traction force is $|F_x| < \mu_{max} F_y(t)$ (Coulomb's relationship).

5.2.5 Orthogonality of the normal modes

An important property of the normal modes, is their orthogonality with respect to the mass and the stiffness matrix. To clarify this property, it is useful to recall eq. (5.25):

$$\left(-\omega^2 \mathbf{M} + \mathbf{K}\right) \mathbf{u} = \mathbf{0} \tag{5.38}$$

this equation can be written for two different modes, namely the $i-th$ and $j-th$ mode:

$$\begin{aligned} \left(\mathbf{K} - \omega_i^2 \mathbf{M}\right) \mathbf{u}^{(i)} &= 0 \\ \left(\mathbf{K} - \omega_j^2 \mathbf{M}\right) \mathbf{u}^{(j)} &= 0 \end{aligned} \tag{5.39}$$

by multiplying on the left the first by $\mathbf{u}^{(j)}$ and the second by $\mathbf{u}^{(i)}$:

$$\begin{cases} \mathbf{u}^{(j)^T} \mathbf{K} \mathbf{u}^{(i)} - \omega_i^2 \mathbf{u}^{(j)^T} \mathbf{M} \mathbf{u}^{(i)} = 0 \\ \\ \mathbf{u}^{(i)^T} \mathbf{K} \mathbf{u}^{(j)} - \omega_j^2 \mathbf{u}^{(i)^T} \mathbf{M} \mathbf{u}^{(j)} = 0 \end{cases} \tag{5.40}$$

By subtracting the first to the second equation:

$$\left(\omega_i^2 - \omega_j^2\right) \mathbf{u}^{(j)^T} \mathbf{M} \mathbf{u}^{(i)} = 0 \tag{5.41}$$

If no eigenvalues has *algebraic multiplicity* MA larger than 1, then $\left(\omega_i^2 - \omega_j^2\right) \neq 0$ if $j \neq i$, the following two orthogonality conditions can be written:

$$\begin{cases} \mathbf{u}^{(j)^T}\mathbf{M}\mathbf{u}^{(i)} = m_i\delta_{ij} \\ \\ \mathbf{u}^{(j)^T}\mathbf{K}\mathbf{u}^{(i)} = \omega_i^2 m_i\delta_{ij} \end{cases} \tag{5.42}$$

where the second is obtained by replacing the first into the first of the (5.40). These relationships are the orthogonality conditions of $\mathbf{u}^{(i)}$ and $\mathbf{u}^{(j)}$ with respect to $\mathbf{M}$ and $\mathbf{K}$, while δ_{ij} is the *Kronecker delta*:

$$\delta_{ij} = \begin{cases} 1 & \text{if} \quad i = j \\ 0 & \text{if} \quad i \neq j \end{cases} \tag{5.43}$$

m_i is the *modal mass* and is the dependent on the representation of the eigenvector (remember that if $\mathbf{u}^{(i)}$ is an eigenvector, also $\alpha\mathbf{u}^{(i)}$ is an eigenvector for any $\alpha \in \mathbb{R}$). The eigenvectors can be normalized to unity modal mass by means of the following substitution:

$$\tilde{\mathbf{u}}^{(i)} = \frac{\mathbf{u}^{(i)}}{\sqrt{m_i}} \qquad i = 1, ...n \tag{5.44}$$

therefore the units of $\tilde{\mathbf{u}}$ are $\frac{1}{[\sqrt{kg}]}$. In the following, all the eigenvectors will be normalized to unity modal mass and the tilde will be omitted: with normalized eigenvectors, the (5.42) becomes:

$$\begin{cases} \mathbf{u}^{(j)^T}\mathbf{M}\mathbf{u}^{(i)} = \delta_{ij} \\ \\ \mathbf{u}^{(j)^T}\mathbf{K}\mathbf{u}^{(i)} = \omega_i^2\delta_{ij} \end{cases} \tag{5.45}$$

or in matrix form:

$$\begin{cases} \mathbf{U}^T\mathbf{M}\mathbf{U} = \mathbf{I} \\ \\ \mathbf{U}^T\mathbf{K}\mathbf{U} = \mathbf{\Lambda} \end{cases} \tag{5.46}$$

where $\mathbf{I}$ is the identity matrix, $\mathbf{\Lambda}$ is a diagonal matrix containing the eigenvalues ω_i^2 on the diagonal, while $\mathbf{U}$ is the *modal matrix*. Each column of $\mathbf{U}$ is one of the n normalized eigenvectors.
In the example of the 1/4 car model, the non-normalized modal matrix is:

$$\mathbf{U} = \begin{bmatrix} \dfrac{2}{\sqrt{7961} - 89} & \dfrac{2}{-\sqrt{7961} - 89} \\ \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 8.9112 & -0.01122 \\ \\ 1 & 1\cdot \end{bmatrix} \tag{5.47}$$

The modal mass m_1 is:

$$
\begin{aligned}
m_1 = \mathbf{u}^{(1)^T}\mathbf{M}\mathbf{u}^{(1)} &= \begin{bmatrix} \dfrac{2}{\sqrt{7961}-89} & 1 \end{bmatrix} \begin{bmatrix} 10m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \dfrac{2}{\sqrt{7961}-89} \\ 1 \end{bmatrix} \\
&= \left(1 + \frac{40}{\left(\sqrt{7961}-89\right)^2}\right) m = 795.1m = 19877.5kg
\end{aligned} \tag{5.48}
$$

in the same way, the modal mass m_2 can be calculated:

$$
m_2 = \mathbf{u}^{(2)^T}\mathbf{M}\mathbf{u}^{(2)} = \left(1 + \frac{40}{\left(-\sqrt{7961}-89\right)^2}\right) m = 1.0012m = 25.0315kg \tag{5.49}
$$

The normalized modal matrix is:

$$
\mathbf{U} = \begin{bmatrix} \dfrac{8.9112}{\sqrt{19877.5}} & \dfrac{-0.01122}{\sqrt{25.0315}} \\ \dfrac{1}{\sqrt{19877.5}} & \dfrac{1}{\sqrt{25.0315}} \end{bmatrix} \frac{1}{\sqrt{kg}} = \begin{bmatrix} 6.32 & -0.22 \\ 0.709 & 19.98 \end{bmatrix} \frac{1}{\sqrt{kg}} \tag{5.50}
$$

5.2.6 Modal expansion theorem

Thanks to the orthogonality of the normal modes, it will be shown that every vector in the configuration space can be written as a linear combination of the eigenvectors, *i.e.* the eigenvectors are a basis of the configuration space. Referring to the eigenvectors normalized at unity normal modes, this property can be stated as follows:

$$
\forall \mathbf{v} \in \mathbb{R}^n \quad , \quad \mathbf{v} = \sum_{r=1}^{n} c_r \mathbf{u}^{(r)} \tag{5.51}
$$

To prove this theorem, we need to find the coefficients c_r, therefore we multiply on the left by $\mathbf{u}^{(j)^T}\mathbf{M}$:

$$
\mathbf{u}^{(j)^T}\mathbf{M}\mathbf{v} = \sum_{r=1}^{n} c_r \mathbf{u}^{(j)^T}\mathbf{M}\mathbf{u}^{(r)} = \sum_{r=1}^{n} c_r \delta_{jr} = c_j \tag{5.52}
$$

In the same way, multiplying on the left by $\mathbf{u}^{(j)^T}\mathbf{K}$:

$$
\mathbf{u}^{(j)^T}\mathbf{K}\mathbf{v} = \sum_{r=1}^{n} c_r \mathbf{u}^{(j)^T}\mathbf{K}\mathbf{u}^{(r)} = \sum_{r=1}^{n} c_r \omega_j^2 \delta_{jr} = c_j \omega_j^2 \tag{5.53}
$$

In matrix form, the theorem is:

$$\begin{cases} \mathbf{v} = \mathbf{U}\mathbf{c} \\ \mathbf{c} = \mathbf{U}^T\mathbf{M}\mathbf{v} \\ \mathbf{\Lambda}\mathbf{c} = \mathbf{U}^T\mathbf{K}\mathbf{v} \end{cases} \tag{5.54}$$

The most signifcant application of this theorem is the transformation of a configuration $q(t)$: in this case the vector of coefficients $\mathbf{c}$ is indicated as $\eta(t)$ and is the vector of the *modal coordinates*.

$$\begin{cases} \mathbf{q}(t) = \mathbf{U}\boldsymbol{\eta}(t) \\ \boldsymbol{\eta}(t) = \mathbf{U}^T\mathbf{M}\mathbf{q}(t) \\ \mathbf{\Lambda}\boldsymbol{\eta}(t) = \mathbf{U}^T\mathbf{K}\mathbf{q}(t) \end{cases} \tag{5.55}$$

From eq. (5.55), it is easy to see that $\mathbf{U}$ is invertible, and its inverse $\mathbf{U}^{-1} = \mathbf{U}^T\mathbf{M}$: by means of $\mathbf{U}$ and $\mathbf{U}^T\mathbf{M}$, the problem of motion can be rewritten from the *configuration space* to the *modal space* and viceversa.

5.2.7 Response of a MDOF system to initial conditions

$$\begin{cases} \mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0} \\ \mathbf{q}\,(0) = \mathbf{q}_0 \\ \dot{\mathbf{q}}\,(0) = \dot{\mathbf{q}}_0 \end{cases} \tag{5.56}$$

Considering a Cauchy problem, the equation of motion can be projected in the modal coordinates using eq. (5.55):

$$\mathbf{M}\mathbf{U}\ddot{\boldsymbol{\eta}}\,(t) + \mathbf{K}\mathbf{U}\boldsymbol{\eta}\,(t) = 0 \tag{5.57}$$

premultiplying by $\mathbf{U}^T\mathbf{M}$:

$$\mathbf{U}^T\mathbf{M}\mathbf{U}\ddot{\boldsymbol{\eta}} + \mathbf{U}^T\mathbf{K}\mathbf{U}\boldsymbol{\eta} = 0 \quad \Rightarrow \quad \mathbf{I}\ddot{\boldsymbol{\eta}} + \mathbf{\Lambda}\boldsymbol{\eta} = 0 \tag{5.58}$$

where orthogonality has been used to reduce the set of n coupled equations to n uncoupled equations:

$$\ddot{\eta}_r(t) + \omega_r^2\eta_r(t) = 0 \quad , \quad r = 1, ...n \tag{5.59}$$

Each equation is equivalent to a 1-DOF system, therefore the solution in the modal space is:

$$\eta_r(t) = C_r\cos(\omega_r t - \psi_r) \quad , \quad r = 1, ...n \tag{5.60}$$

This solution is to be projected back to the configuration space using eq. (5.55):

$$
\begin{aligned}
q_i(t) &= \sum_{r=1}^{n} U_{ir}\eta_r(t) = \sum_{r=1}^{n} u_i^{(r)}\eta_r(t) = \\
&= \sum_{r=1}^{n} u_i^{(r)} C_r \cos(\omega_r t - \psi_r) \quad , \quad i = 1, ...n
\end{aligned}
\tag{5.61}
$$

The initial conditions are to be projected on the modal basis using again eq. (5.55):

$$
\begin{cases}
\mathbf{u}^{(r)^T}\mathbf{M}\mathbf{q}_0 = \eta_r(0) = C_r \cos(\psi_r) \\
\mathbf{u}^{(r)^T}\mathbf{M}\dot{\mathbf{q}}_0 = \eta_r(0) = C_r\omega_r \sin(\psi_r)
\end{cases}
$$

$$
\begin{cases}
C_r = \sqrt{\left(\mathbf{u}^{(r)^T}\mathbf{M}\mathbf{q}_0\right)^2 + \left(\dfrac{\mathbf{u}^{(r)^T}\mathbf{M}\dot{\mathbf{q}}_0}{\omega_r}\right)^2} \\
tg(\psi_r) = \dfrac{\mathbf{u}^{(r)^T}\mathbf{M}\dot{\mathbf{q}}_0}{\omega_r \mathbf{u}^{(r)^T}\mathbf{M}\mathbf{q}_0}
\end{cases}
\tag{5.62}
$$

The values of C_r and ψ_r are dependent upon the initial conditions only; as shown in fig. 5.1, each modal coordinate oscillates with constant mechanical energy. The most important conclusion of this analysis is that during free oscillations there is no energy transfer among different modes.

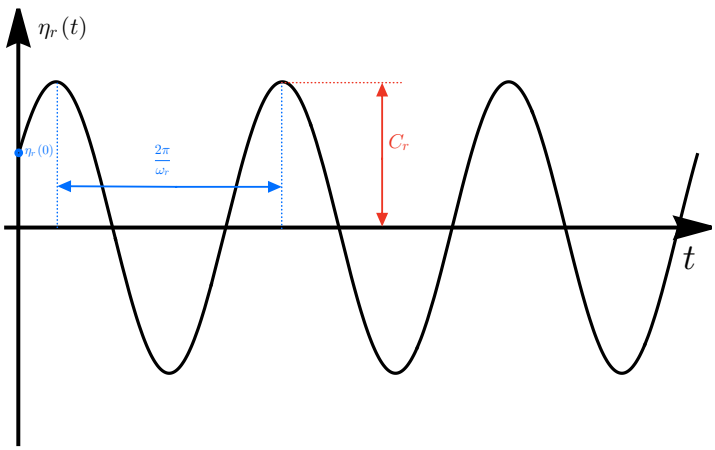

Figure 5.2: Response to initial conditions of a MDOF system: modal coordinate $\eta_r(t)$

5.2.8 Response of a MDOF linear system to an arbitrary excitation

Considering a MDOF system with arbitrary excitation $\mathbf{f}(t)$:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{f}\left(t\right) \tag{5.63}$$

the equation of motion can be projected in the modal coordinates using eq. (5.55) and multiplying on the left by $\mathbf{U}^T$:

$$\mathbf{U}^T\mathbf{M}\mathbf{U}\ddot{\boldsymbol{\eta}} + \mathbf{U}^T\mathbf{K}\mathbf{U}\boldsymbol{\eta} = \mathbf{U}^T\mathbf{f}\left(t\right) \tag{5.64}$$

It is useful to define the vector of the *modal force*: $\mathbf{p}(t) = \mathbf{U}^T\mathbf{f}\left(t\right)$. Thanks to mode orthogonality, the n equations are decoupled:

$$\mathbf{I}\ddot{\boldsymbol{\eta}} + \boldsymbol{\Lambda}\boldsymbol{\eta} = \mathbf{p}(t) \tag{5.65}$$

the $i-th$ scalar equation is:

$$\ddot{\eta}_i + \omega_i^2\eta_i = p_i\left(t\right) = \sum_{j=1}^{n} U_{ji}f_j = \sum_{j=1}^{n} u_j^{(i)}f_j \tag{5.66}$$

It is clear that each $\eta_i(t)$ is governed by the equation of a 1-DOF system with arbitrary excitation, therefore the response is obtained by means of the convolution integral:

$$\eta_i\left(t\right) = \int_0^t p_i\left(\tau\right)h_i\left(t-\tau\right)d\tau \quad \text{with} \quad h_i\left(t\right) = \frac{1}{\omega_i}\sin\left(\omega_i t\right) \tag{5.67}$$

5.2.9 Frequency Response Function (FRF) of a MDOF linear system

We consider again a generic MDOF undamped linear system with arbitrary excitation $\mathbf{f}(t)$:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{f}\left(t\right) \tag{5.68}$$

From section 5.2.8, we know that the equations can be decoupled using the modal expansion theorem, so that we get again eq. (5.66):

$$\ddot{\eta}_i + \omega_i^2\eta_i = p_i\left(t\right) = \sum_{j=1}^{n} U_{ji}f_j = \sum_{j=1}^{n} u_j^{(i)}f_j \tag{5.69}$$

We can apply the Fourier transform to both sides of the equation:

$$-\omega^2\xi_i + \omega_i^2\xi_i = P_i \quad \Rightarrow \quad \xi_i = \frac{P_i}{-\omega^2 + \omega_i^2} \tag{5.70}$$

where $\xi_i(\omega)$ is the Fourier transform of $q_i(t)$ and $P_i(\omega)$ is the Fourier transform of the $i-th$ component of the modal force $p_i(t)$.
The modal expansion theorem $\mathbf{q}(t) = \mathbf{U}\boldsymbol{\eta}(t)$ can be Fourier transformed to move back to the generalized coordinate $Q_r(\omega)$ (which is the Fourier transform of $q_r(t)$):

$$\begin{aligned} Q_r(\omega) &= \sum_{i=1}^{n} u_r^{(i)}\xi_i = \sum_{i=1}^{n} u_r^{(i)} \frac{\sum_{j=1}^{n} u_j^{(i)} F_j(\omega)}{-\omega^2 + \omega_i^2} = \\ &= \sum_{j=1}^{n} \left[\sum_{i=1}^{n} \frac{u_r^{(i)} u_j^{(i)}}{-\omega^2 + \omega_i^2} \right] F_j(\omega) = \sum_{j=1}^{n} H_{rj}(\omega) F_j(\omega) \end{aligned} \tag{5.71}$$

From eq. (5.71) it is pointed out the relationship between the Frequency Response Function $H_{rj}(\omega)$ and the modal matrix $\mathbf{U}$:

$$H_{rj}(\omega) = \sum_{i=1}^{n} \frac{u_r^{(i)} u_j^{(i)}}{-\omega^2 + \omega_i^2} \tag{5.72}$$

the numerator of the $i-th$ term in the (5.73) is the product of two components of the $i-th$ eigenvector, therefore if is called *modal constant*. At the denominator, the term $-\omega^2 + \omega_i^2$ is zero if $\omega = \omega_i$, *i.e.* at the natural frequencies, therefore a vertical asymptote is seen at each natural frequency. Close to each natural frequency the $i-th$ term is dominant and the FRF can be approximated by the $i-th$ term:

$$H_{rj}(\omega) \cong \frac{u_r^{(i)} u_j^{(i)}}{-\omega^2 + \omega_i^2} \quad \text{for} \quad \omega \approx \omega_i \tag{5.73}$$

for lightly damped systems, the half power points approach described in section 4.4.2 can be used close to each $i-th$ resonance to characterize the value of the modal damping ζ_i.
In fig. 5.3 the FRFs for the quarter car model described in section 5.2.1 can be seen: H_{11} and $H22$ are found along the diagonal of the $\mathbf{H}$ matrix and it can be proven that for the component on the diagonal two resonances are always separated by an *antiresonance*, that is to say a zero of the FRF. H_{12} is a component out of the diagonal of the $\mathbf{H}$ matrix and it does not have an antiresonance. Note that from eq. (5.73) it is easy to see that the $\mathbf{H}$ matrix is diagonal, therefore $H_{21} = H_{12}$.

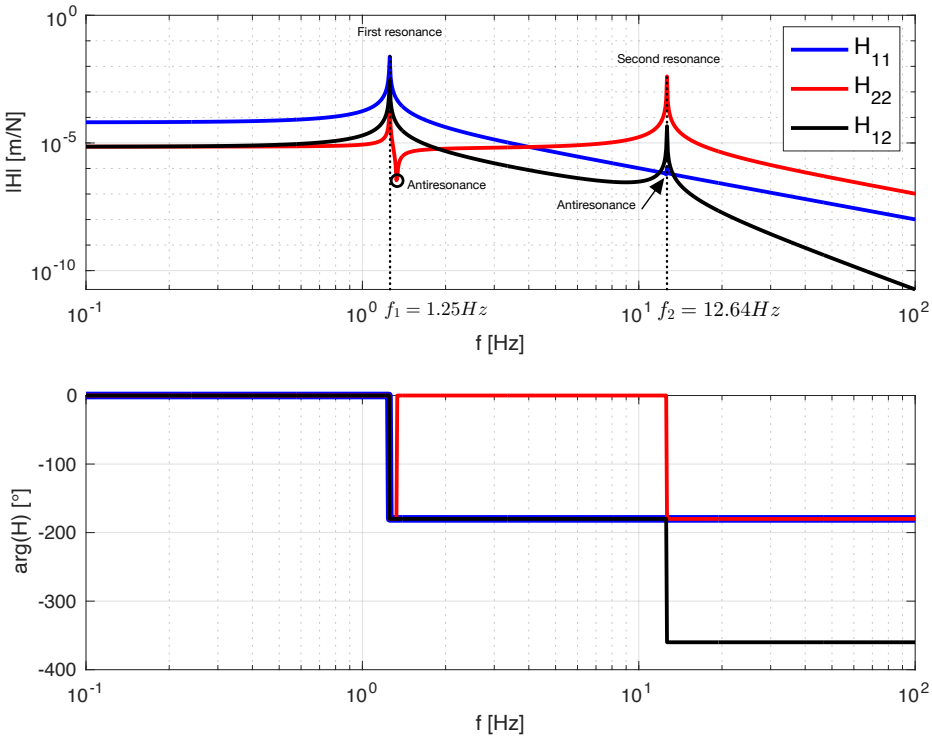

Figure 5.3: FRFs of the undamped quarter car model described in section 5.2.1

5.3 Damped MDOF systems

5.3.1 MDOF systems with proportional damping

If the MDOF linear system to be studied is characterized by viscous damping:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{f} \tag{5.74}$$

then, in general, the modal expansion theorem stated in eq. (5.66) cannot be used anymore to decouple the n equations of motion. In order to be able to decouple the equations with the expansion theorem it is necessary to use a simplified approach, called *proportional damping*. Proportional damping consists in selecting a matrix $\mathbf{C}$ that can be written in the following form:

$$\mathbf{C} = \alpha\mathbf{M} + \beta\mathbf{K} \tag{5.75}$$

where α and β are dimensional coefficients called *Rayleigh damping parameters*. Therefore, using this approach, the matrix $\mathbf{C}$ is a linear combination of $\mathbf{M}$ and $\mathbf{K}$ and it can be decoupled using the following approach:

$$\begin{gathered}\mathbf{U}^T\mathbf{M}\mathbf{U}\ddot{\boldsymbol{\eta}} + \alpha\mathbf{U}^T\mathbf{M}\mathbf{U}\dot{\boldsymbol{\eta}} + \beta\mathbf{U}^T\mathbf{K}\mathbf{U}\dot{\boldsymbol{\eta}} + \mathbf{U}^T\mathbf{K}\mathbf{U}\boldsymbol{\eta} = \mathbf{U}^T\mathbf{f} \\ \mathbf{I}\ddot{\boldsymbol{\eta}} + \alpha\mathbf{I}\dot{\boldsymbol{\eta}} + \beta\boldsymbol{\Lambda}\dot{\boldsymbol{\eta}} + \boldsymbol{\Lambda}\boldsymbol{\eta} = \mathbf{U}^T\mathbf{f}\end{gathered} \tag{5.76}$$

the $i-th$ decoupled equation is:

$$\ddot{\eta}_i + \alpha\dot{\eta}_i + \beta\omega_i^2\dot{\eta}_i + \omega_i\eta_i = p_i \tag{5.77}$$

this equation can be compared to that of a 1-DOF system with viscous damping and external forcing as follows:

$$\begin{gathered}\ddot{\eta}_i + \left(\alpha + \beta\omega_i^2\right)\dot{\eta}_i + \omega_i\eta_i = p_i \\ \ddot{\eta}_i + \left(2\zeta_i\omega_i^2\right)\dot{\eta}_i + \omega_i\eta_i = p_i\end{gathered} \tag{5.78}$$

therefore the *modal damping* ζ_i can be evaluated from α, β and ω_i by imposing that the two formulations are equivalent:

$$2\zeta_i\omega_i = \alpha + \beta\omega_i^2 \quad \Rightarrow \quad \zeta_i = \frac{1}{2}\left(\frac{\alpha}{\omega_i} + \beta\omega_i\right) \quad , \quad i = 1, -n \tag{5.79}$$

from this relationship it is clear that the contribution $\alpha\mathbf{M}$ is dominant al low frequency, while $\beta\mathbf{K}$ is dominant al high frequency. In order to select the Rayleigh parameters α and β that can approximate damping in the best way, the following approach is used:

1. a suitable modal damping value $\bar{\zeta}$ is chosen either from experimental measurements or by experience (note that modal damping is different for each natural frequency, nonetheless here only one value can be selected);
2. a lower frequency ω_1 is chosen (*e.g.* ω_1 can be lower than the first natural frequency)
3. a higher frequency ω_2 is chosen (*e.g.* ω_2 can be higher than the maximum significant frequency in the force $\mathbf{f}(t)$).
4. the function $\zeta_i(\omega_i)$ in (5.79) is forced to pass in the points $(\omega_1, \bar{\zeta}$ and $(\omega_2, \bar{\zeta}$.

The last condition gives the following set of two linear equations:

$$\begin{cases} \bar{\zeta} = \dfrac{1}{2}\left(\dfrac{\alpha}{\omega_1} + \beta\omega_1\right) \\ \bar{\zeta} = \dfrac{1}{2}\left(\dfrac{\alpha}{\omega_2} + \beta\omega_2\right) \end{cases} \quad \Rightarrow \quad \begin{cases} \alpha = \dfrac{2\omega_1\omega_2}{\omega_1 + \omega_2}\bar{\zeta} \\ \beta = \dfrac{2}{\omega_1 + \omega_2}\bar{\zeta} \end{cases} \tag{5.80}$$

If proportional damping is used, the same modal matrix $\mathbf{U}$ of the undamped problem can decouple the equations of the damped problem and the natural frequencies are unchanged. Due to its simple application, this approach is frequently used in commercial software for finite element analysis; the main drawbacks are:

- that the modes are completely wrong if the damping is essentially non-proportional (obviously, problems with large damping will be more affected by this issue than problems with small damping);
- that the value of damping is very high out of the interval (ω_1, ω_2), so it is of paramount importance to avoid extrapolating out of that interval.

Finally, by Fourier transforming the second of eq. (5.78) and using the same approach seen in section 5.2.9, the FRF for a MDOF system with proportional damping can be obtained:

$$H_{rj}(\omega) = \sum_{i=1}^{n} \frac{u_r^{(i)} u_j^{(i)}}{-\omega^2 + \omega_i^2 + 2i\zeta_i\omega_i\omega} \tag{5.81}$$

and close to the $i-th$ natural frequency this equation can be approximated as follows:

$$H_{rj}(\omega) \cong \frac{u_r^{(i)} u_j^{(i)}}{-\omega^2 + \omega_i^2 + 2i\zeta_i\omega_i\omega} \quad \text{for} \quad \omega \approx \omega_i \tag{5.82}$$

Note that the FRF is a complex function of the circular frequency ω (or equivalently of the frequency f), while it was a real function for undamped systems. If the system has a small damping, we can conclude that:

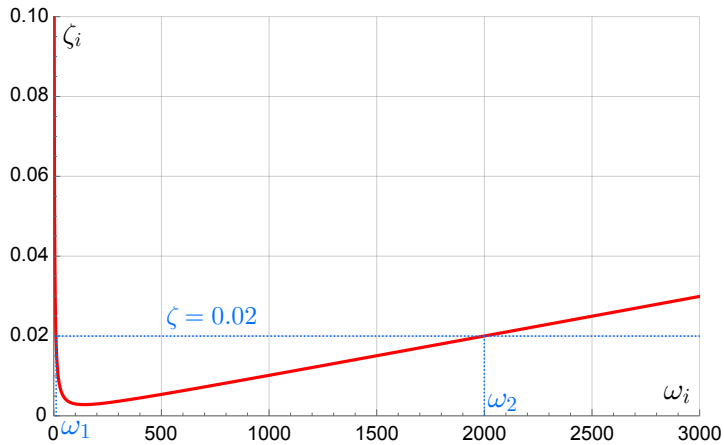

Figure 5.4: Choice of the Rayleigh parameter ($\bar{\zeta} = 0.02$, $\omega_1 = 10$, $\omega_2 = 2000$

- close to the $i - th$ natural frequency, the $i - th$ mode is dominant and the other modes are neglibile;
- $\omega = \omega_i$ is equivalent to the resonance condition;
- H_{rj} is purely imaginary for $\omega = \omega_i$

As an example, we can consider the quarter car model described in section 5.2.1 if we assume that a proportional damping is added such that for the two natural frequencies $\zeta_1 = \zeta_2 = 0.1$, the FRFs are shown in fig. 5.5. Note that this model is unphysical, since damping in the tyres is usually negligible with respect to damping in the car shock absorber, conversely any model with proportional damping will introduce an amount of damping on the tyres.

5.3.2 MDOF systems with non proportional damping

The MDOF linear system under investigation has viscous damping:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{f} \tag{5.83}$$

to study the problem without introducing an approximation, it is necessary to reduce to the first order as described in section 5.1.2. In this problem, order reduction can be performed using block matrices. The new unknown is $\mathbf{x} = \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix}$ and an identity

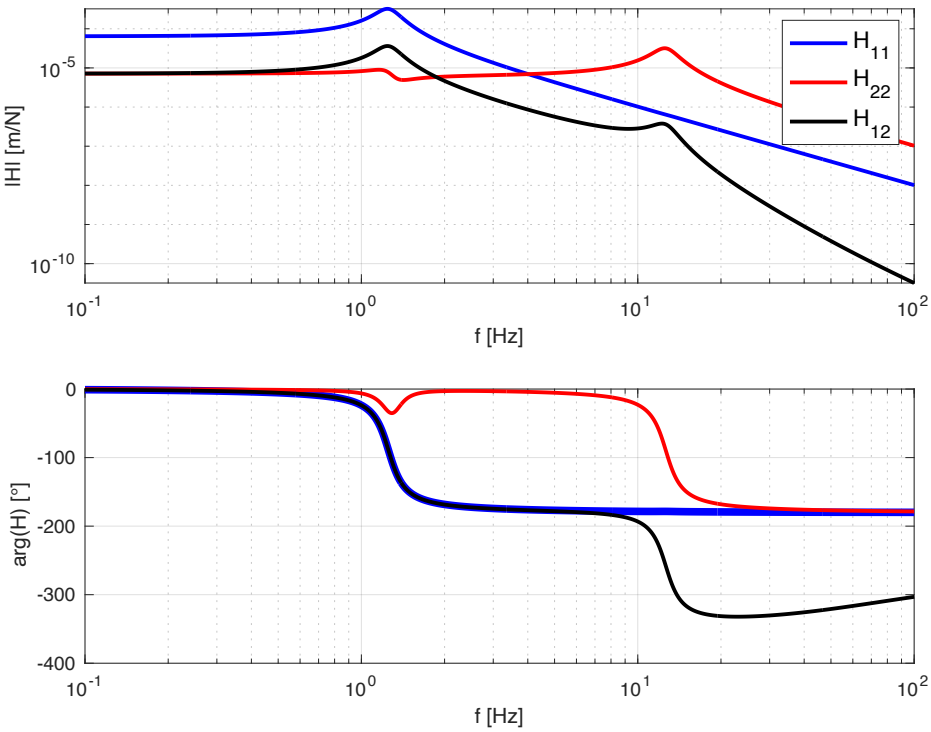

Figure 5.5: FRFs of the quarter car model described in section 5.2.1 with non-physical proportional damping ($\zeta_1 = \zeta_2 = 0.1$)

is added to the equation (5.83):

$$\begin{cases} \mathbf{M}\dot{\mathbf{x}}_2 + \mathbf{C}\dot{\mathbf{x}}_1 + \mathbf{K}\mathbf{x}_1 = \mathbf{f} \\ \mathbf{M}\dot{\mathbf{x}}_1 - \mathbf{M}\mathbf{x}_2 = \mathbf{0} \end{cases} \tag{5.84}$$

using block matrices, this set of $2n$ equation becomes

$$\begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{x}}_1 \\ \dot{\mathbf{x}}_2 \end{bmatrix} + \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \tag{5.85}$$

The set of equations is in the form:

$$\mathbf{A}\dot{\mathbf{x}} + \mathbf{B}\mathbf{x} = \bar{\mathbf{f}} \tag{5.86}$$

multiplying (5.86) by $\mathbf{A}^{-1}$ on the left side, the equation can be placed in *canonical form*, which is very useful if it has to be solved numerically, *e.g.* using Matlab and ODE45:

$$\dot{\mathbf{x}} = -\mathbf{A}^{-1}\mathbf{B}\mathbf{x} + \mathbf{A}^{-1}\bar{\mathbf{f}} \tag{5.87}$$

For this problem, it is possible to find the eigenvalues and the eigenvectors that decouple the set of equations in the state space, rather than in the configuration space. Referring to the free problem:

$$\mathbf{A}\dot{\mathbf{x}} + \mathbf{B}\mathbf{x} = \mathbf{0} \tag{5.88}$$

this is a 1st order linear system, therefore we look for an exponential solution in the form:

$$\mathbf{x}(t) = \boldsymbol{\phi} e^{\lambda t} = \begin{bmatrix} \boldsymbol{\psi} \\ \lambda\boldsymbol{\psi} \end{bmatrix} e^{\lambda t} \tag{5.89}$$

where $\boldsymbol{\phi}$ is an eigenvector having size $2n$, which can be decomposed into the vector $\boldsymbol{\psi}$, which corresponds to the n generalized coordinates, and $\lambda\boldsymbol{\psi}$, which corresponds to the n generalized velocities.
Replacing eq. (5.89) into (5.88) :

$$(\lambda\mathbf{A} + \mathbf{B})\,\boldsymbol{\phi} = \mathbf{0} \quad \Rightarrow \quad \left(\lambda\mathbf{I} + \mathbf{A}^{-1}\mathbf{B}\right)\boldsymbol{\phi} = \mathbf{0} \tag{5.90}$$

imposing that $\mathbf{A}^{-1}\mathbf{B}$ is singular, $2n$ eigenvalues and eigenvector couples $(\lambda_r, \boldsymbol{\phi}_r)$ can be found. These eigenvalues and eigenvectors must be complex and conjugate in pairs, since $\mathbf{x}(t)$ must be real, therefore if the $r-th$ couple is $(\lambda_r, \boldsymbol{\phi}_r)$, then the couple $(\lambda_r^*, \boldsymbol{\phi}_r^*)$ is certainly found among eigenvalues/eigenvectors.

The real vibration associated to the couples $(\lambda_r, \boldsymbol{\phi}_r)$ and $(\lambda_r^*, \boldsymbol{\phi}_r^*)$ is given by the summation:

$$\mathbf{x}^{(r)}(t) = \boldsymbol{\phi}^{(r)} e^{\lambda_r t} + \boldsymbol{\phi}^{(r)^*} e^{\lambda_r^* t} =$$

$$= 2Re\left[\boldsymbol{\phi}^{(r)} e^{\lambda_r t}\right] =$$

(5.91)

$$= 2Re\left[\left(\boldsymbol{\phi}_R^{(r)} + i\boldsymbol{\phi}_I^{(r)}\right) e^{\lambda_r^R t} e^{i\lambda_r^I t}\right] =$$

$$= 2e^{\lambda_r^R t}\left(\boldsymbol{\phi}_R^{(r)} \cos\left(\lambda_r^I t\right) - \boldsymbol{\phi}_I^{(r)} \sin\left(\lambda_r^I t\right)\right)$$

This vibration gives rise to non synchronous motion, *i.e.* if the $r-th$ mode is excited, the states $x_i(t)$ are oscillating functions that are not in phase with respect to each other. This point is clarified by fig. 5.6, where the cosine and sine function are summated with 1 and $\frac{1}{2}$ coefficient respectively; if they are summated with different coefficients, which will certainly occur for another component, the response will have a different shape than the one shown in the red curve. If the modal damping of the

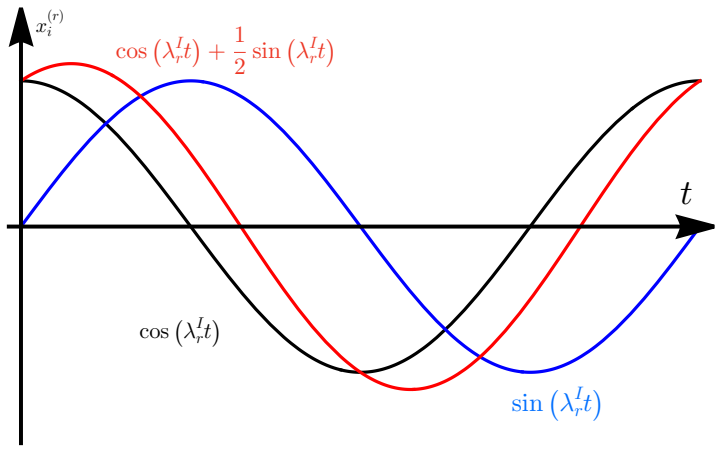

Figure 5.6: Example of response of the $r-th$ complex mode: the $i-th$ component of $\mathbf{x}^{(r)}(t)$ depends upon the coefficients of the sine and cosine terms

$r-th$ complex mode is subcritical, a correspondence can be drawn between the real response associated to the mode and the response of a damped 1-DOF system eq. (4.30); in particular, the eigenvalues λ_r and λ_r^* are related to the complex roots:

(5.92) $$\lambda_r = -\zeta_r \omega_r \pm i\omega_r\sqrt{1-\zeta_r^2}$$

therefore the $r-th$ natural frequency and the modal damping are:

$$\lambda_r = -\zeta_r \omega_r \pm i\omega_r \sqrt{1-\zeta_r^2} \tag{5.93}$$

We can conclude that the modulus of the eigenvalue is the natural frequency, while the real part of the eigenvalue allows for calculating the modal damping ζ_r. Note that the real part of all the eigenvalues must be real or zero, otherwise the equilibrium point would be unstable.

$$\begin{cases} |\lambda_r| = \sqrt{(\zeta_r \omega_r)^2 + \omega_r^2(1-\zeta_r^2)} = \omega_r \\ \\ Re\,(\lambda_r) = -\zeta_r \omega_r \quad \Rightarrow \quad \zeta_r = -\dfrac{Re\,(\lambda_r)}{\omega_r} = -\dfrac{Re\,(\lambda_r)}{|\delta_r|} \end{cases} \tag{5.94}$$